

“ IF YOU CAN BRING A **MOVABLE THRESHING MACHINE...** IT WILL BE AMONG THE **MOST VALUABLE INSTITUTIONS** IN THIS COUNTRY. ”

George Washington, first US president, from a letter to Thomas Jefferson demonstrating the importance of the threshing machine, 1786



THE SENSATION OF 1783

was the first manned flight, in a hot-air balloon on November 21, in Paris. Piloted by Pilâtre de Rozier (1754–85) and the Marquis d'Arlandes, the balloon was the invention of French papermakers the **Montgolfier brothers**, Joseph-Michel (1740–1810) and Étienne (1745–99). The Montgolfiers had previously discovered that when hot air is trapped inside a bag it causes the bag to float upward, since hot air is less dense than cold air.

Ten days later, French physicist **Jacques Charles** (1746–1823) and craftsman **Noël Robert** (1760–1820) flew over Paris in

a **balloon filled with hydrogen gas**. They achieved much longer, more controlled flights than with hot air (which quickly cooled).

Also in 1783, Spanish brothers **José** (1754–96) and **Fausto Elhuyar** (1755–1833) isolated

2,953 FEET
THE HEIGHT REACHED BY MONTGOLFIER BALLOON IN 1783

a new metal element by reacting charcoal with tungstic acid. It was later named tungsten, one of the toughest of all metals.

In 1784, English chemist **Henry Cavendish** (1731–1810) showed that water is composed of two gases, hydrogen and oxygen.

Astronomer **John Michell** predicted the **existence of black holes** when he observed that a

star of sufficient mass would have such a strong gravitational field that light could not escape.

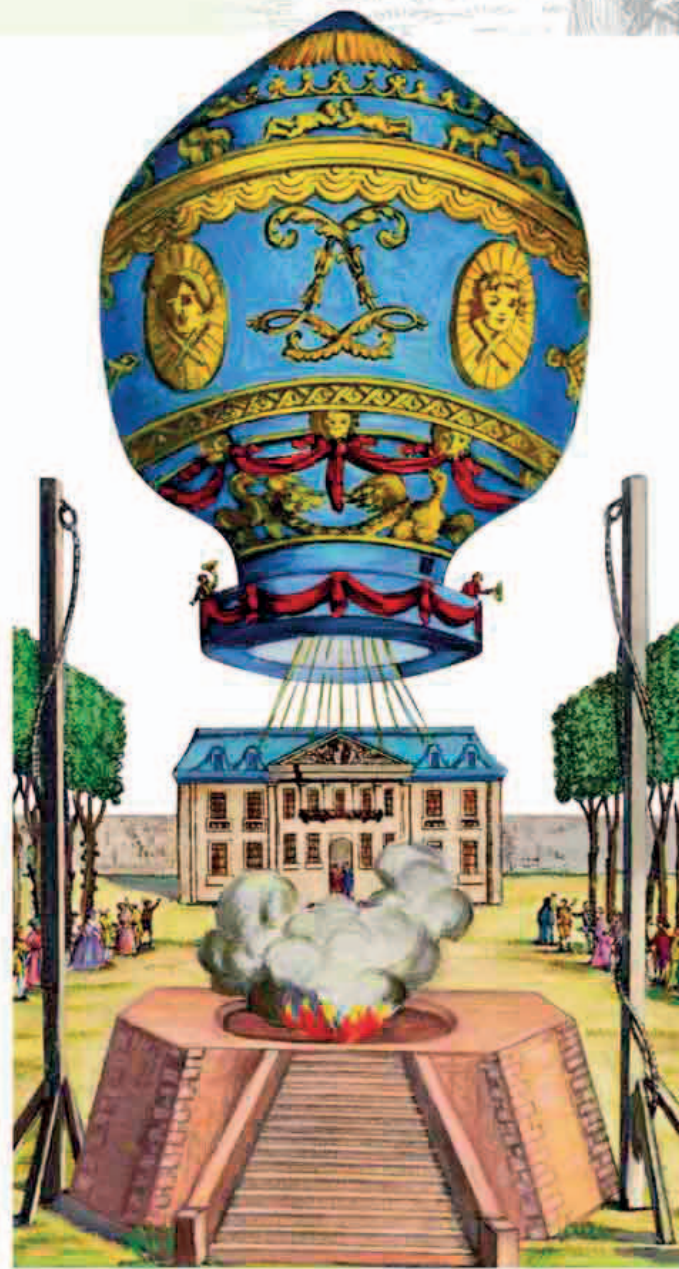
In London, inventor **Joseph Bramah** (1748–1814) patented his **unpickable security lock**, which opened only when the key moved a unique combination of sliders into grooves.

The **biggest question for geologists** in the 1780s was how marine fossils ended up in rocks high in mountains. German geologist Abraham Werner (1749–1817) believed that a single catastrophe, such as an almighty flood, had reshaped the world dramatically (see 1775–76). But Scottish geologist **James Hutton** (1726–97) did not agree. In his **groundbreaking Theory of the Earth**, published in 1785, he suggested that landscapes are shaped slowly and continually over long periods by repeating cycles of erosion, sedimentation, and uplift. Hutton's theory (now proved correct) meant that Earth must be millions of years old—not just thousands, as was thought at the time

In France, physicist **Charles de Coulomb** (1736–1806) established an important law

First balloon ascent

In this early 19th-century print, Pilâtre de Rozier and the Marquis d'Arlandes acknowledge spectators as they ascend in their Montgolfier hot-air balloon in November 1783.



expansion chamber for gases

brass casing

inlet for gas

Henry Cavendish's eudiometer
This eudiometer is a replica of one used by Cavendish to measure the volume of gases, such as oxygen, in air, produced in a reaction.

November 21, 1783
First manned flight in the Montgolfier hot-air balloon takes place in Paris

December 1, 1783
First flight in a hydrogen gas balloon takes place in Paris

January 15, 1784
Henry Cavendish shows that water comprises hydrogen and oxygen

August 21, 1784 Joseph Bramah patents his security lock

1785 Charles de Coulomb establishes his Law of electrical charges

1785 Andrew Meikle invents the threshing machine

1785 James Hutton publishes *Theory of the Earth*, suggesting slow shaping of landscapes

1785 Jean Blanchard invents the parachute

1785 Edmund Cartwright develops the power loom